



AROR UNIVERSITY OF ART, ARCHITECTURE, DESIGN & HERITAGE, SUKKUR, SINDH

Course: Applied Physics

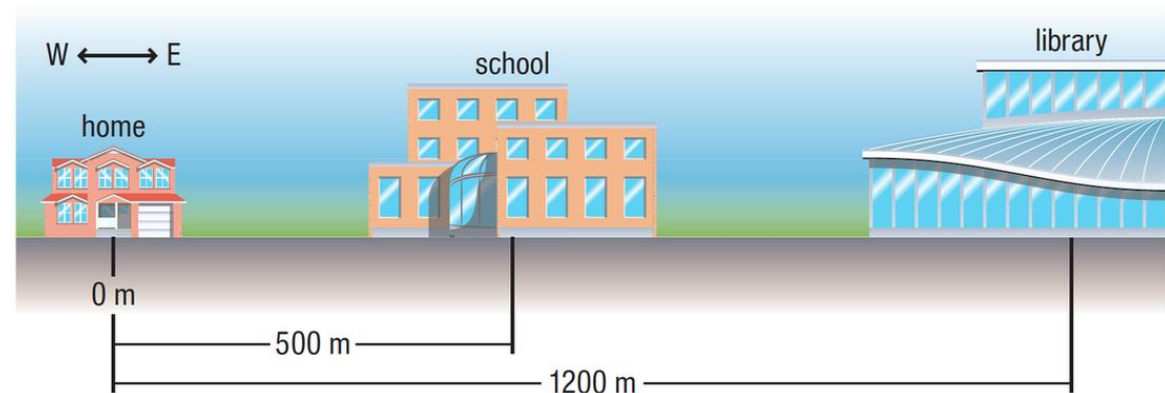
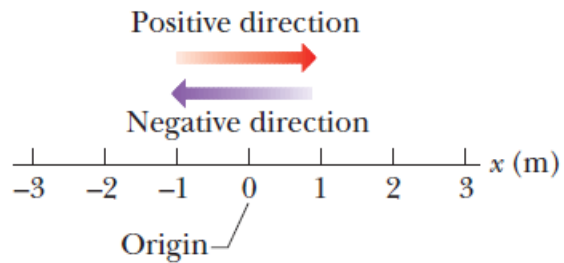
Lecture 03

Instructor:

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1. Position

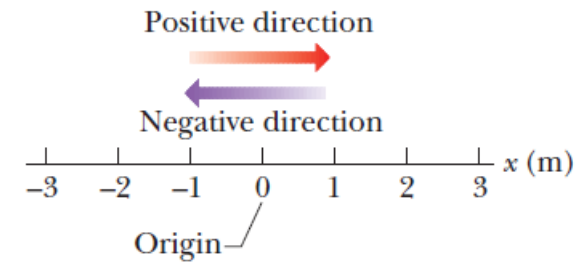
“Position of a body is its location in space relative to a chosen reference point (origin) at a given time.”



2. Displacement

*“A change from position x_1 to position x_2 is called a **displacement** Δx ”*

$$\Delta x = x_2 - x_1.$$



- (The symbol Δ , the Greek uppercase delta, represents a change in a quantity, and it means the final value of that quantity minus the initial value.)
- Displacement is an example of a **vector quantity**, which is a quantity that has both a direction and a magnitude.

Average Speed

Speed → the distance traveled by an object per unit time.

$$\text{Speed} = \frac{\text{Distance}}{\text{Time}}$$

Example:

If a car travels 100 m in 5 s,

$$\text{Speed} = 20 \text{ m/s}$$

Average speed → the total distance traveled divided by the total time taken.

$$\text{Average Speed} = \frac{\text{Total Distance}}{\text{Total Time}}$$

Example:

If a person travels 40 m in 4 s and then 60 m in 6 s,

$$\text{Average Speed} = 10 \text{ m/s}$$

Short summary

- **Speed** → distance per unit time
- **Average speed** → total distance ÷ total time

Average Velocity

Velocity: How fast an object is moving *and* in which direction, at a given moment or over a motion.

Velocity = speed with direction



Example

- A car moving at **60 km/h east**
- A ball falling at **5 m/s downward**

If you only say **60 km/h**, that's speed.

When you add **east**, it becomes velocity.

Basic formula

For straight-line motion:

$$\text{Velocity} = \frac{\text{displacement}}{\text{time}}$$

Average Velocity (2)

Average velocity tells you how fast *and in which direction* an object's position changes over a period of time.

Average velocity =
total change in position ÷ **total time taken**

- Velocity depends on displacement, not distance
- Displacement means **start point** → **end point**, including direction

Simple example

Imagine this:

- You walk **10 m east**
- Then you walk **10 m west**
- Total time = 20 seconds

Distance traveled = 20 m

Displacement = 0 m (you ended where you started)

So,

$$\text{Average velocity} = \frac{0}{20} = 0 \text{ m/s}$$

➡ Even though you were moving the whole time, your **average velocity is zero** because your final position didn't change.

Average Velocity (3)

Another everyday example

- A car travels **100 km north** in **2 hours**

$$\text{Average velocity} = \frac{100 \text{ km north}}{2 \text{ h}} = 50 \text{ km/h north}$$

Notice:

- Speed alone would be **50 km/h**
- Velocity is **50 km/h north** (direction matters)

Average velocity tells how quickly and in what direction an object changes its position overall — not how much ground it covered.

Acceleration

Velocity vs Acceleration

- **Velocity** → tells how fast an object is moving and in which direction
- **Acceleration** → tells how fast the velocity is changing

If velocity is constant → acceleration = 0

If velocity changes → acceleration is present

Acceleration → the rate of change of velocity per unit time.

$$\text{Acceleration} = \frac{\text{Change in velocity}}{\text{Time}}$$

Velocity changes when:

1. Speed increases
2. Speed decreases
3. Direction changes

If any of these occur → acceleration exists.

Example

A car increases speed from 10 m/s to 20 m/s in 5 s

$$\text{Acceleration} = \frac{20 - 10}{5} = 2 \text{ m/s}^2$$

Acceleration (2)

- When a particle's velocity changes, the particle is said to undergo **acceleration** (or to accelerate).
- For motion along an axis, the **average acceleration** a_{avg} over a time interval Δt is:

$$a_{avg} = \frac{v_2 - v_1}{t_2 - t_1} = \frac{\Delta v}{\Delta t},$$

- where the particle has velocity v_1 at time t_1 and then velocity v_2 at time t_2
- The **instantaneous acceleration** (or simply **acceleration**) is:

$$a = \frac{dv}{dt}.$$

Easy Real-Life Understanding

- Car moving at constant speed in a straight line → velocity present, acceleration = 0
- Car speeding up or slowing down → acceleration present
- Car turning at same speed → acceleration present

What happens when a car turns?

- Speed → **same**
- Direction → **changes**

Since direction changes → **velocity changes**

And when velocity changes → **acceleration is present**

Acceleration (3)

Units of acceleration

SI unit of acceleration $\rightarrow \text{m/s}^2$ (metre per second squared)

Explanation

Acceleration means **change in velocity per unit time**.

- Velocity unit $\rightarrow \text{m/s}$
- Time unit $\rightarrow \text{s}$

So,

$$\text{Acceleration unit} = \frac{\text{m/s}}{\text{s}} = \text{m/s}^2$$

g Units.

Large accelerations are sometimes expressed in terms of g units, with

$$1g = 9.8 \text{ m/s}^2 \quad (g \text{ unit}).$$

$g \rightarrow$ a unit of acceleration equal to the **acceleration due to gravity of the Earth**.

$$1g \approx 9.8 \text{ m/s}^2$$

Free fall acceleration

What is free fall?

Free fall → When an object moves **downward only because of gravity**, with **no other forces** like air resistance slowing it down.

Examples:

- Dropping a stone from your hand
- An apple falling from a tree
- A ball dropped from a roof

💡 Important: If air pushes back (like a feather in air), it's **not pure free fall**.

Free fall acceleration?

- The rate at which an object **speeds up while falling**, due to gravity alone.
- If you tossed an object either up or down and could somehow eliminate the effects of air on its flight, you would find that the object accelerates downward at a certain constant rate.
- That rate is called the **free-fall acceleration**, and its magnitude is represented by g .
- The acceleration is independent of the object's characteristics, such as mass, density, or shape; it is the same for all objects.

Free fall acceleration (2)

On Earth, **all objects accelerate at the same rate**, no matter their weight or size

This acceleration = **$g \approx 9.8 \text{ m/s}^2$**

Meaning:

Every second, the speed of a falling object **increases by 9.8 m/s** (if air resistance is ignored).

Simple definition: Free fall acceleration is the acceleration of an object due to gravity alone, causing it to speed up downward at about 9.8 m/s^2 , regardless of its mass.

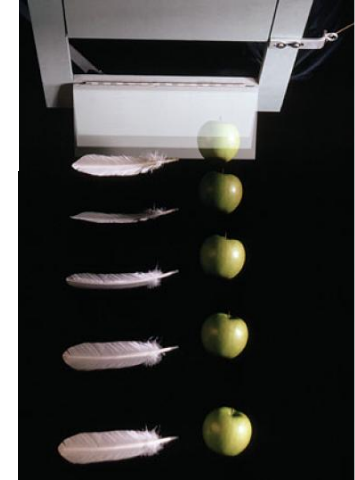


Figure: A feather and an apple free fall in vacuum at the same magnitude of acceleration g .

- The acceleration increases the distance between successive images.
- In the absence of air, the feather and apple fall together.”